

schedule_feature_research

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schedule_feature_research.sh – § . . FEATURE research-scheduling engine

Field	Value
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Script	constitution/scripts/feature/schedule_feature_research.sh
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Overview

schedule_feature_research.sh turns a plain-language FEATURE DESCRIPTION — delivered by the §11.4.140 registered action FEATURE , or by any other caller — into a **SCHEDULED** (never synchronously executed) deep, enterprise-grade research + implementation-planning effort.

feature description

```
|-(1) CREATE → a Type=Task workable item, by DELEGATING to the SAME  
|               §11.4.202 report_item.sh engine (never a duplicate  
|               implementation) – its description embeds the full  
|               research-and-planning WORK PROGRAM as acceptance
```

criteria

```
|-(2) SYNC + PUSH → delegated to report_item.sh exactly as §11.4.202  
|               performs them (absent credentials/client ⇒ honest SKIP,  
|               NEVER a faked push)
```

```
└-(3) QUEUE → append a row to the durable docs/requests/
```

feature_queue.md

(mirrors the §11.4.140 BACKGROUND durable-queue pattern)

The multi-day research / planning / documentation work described INSIDE the item is **not performed by this engine**. It is performed LATER by the project's standing autonomous loop (§11.4.87 / §11.4.94 / §11.4.97 / §11.4.103 / §11.4.126) when it claims the item, driven to a genuinely COMPLETED-and-wired or explicitly evidence-backed CLOSED terminal state under the §11.4.197 research-completion mandate — a FEATURE item is NEVER left sitting un-wired in the backlog.

Prerequisites

- `python3` (+PyYAML) — reads the consumer config and parses the delegate's JSON result.
- The sibling `constitution/scripts/reporting/report_item.sh` engine (§11.4.202) — this engine DELEGATES to it rather than reimplementing item-creation / DB-sync / tracker-push.
- A **consumer-owned config** (the engine refuses to guess — §11.4.6):

```
mkdir -p .helix
```

```
cp constitution/scripts/feature/feature.example.yaml .helix/feature.yaml
```

```
$EDITOR .helix/feature.yaml # db, id_prefix, research_doc_root, sync_command, trackers
```

Search order: `$HELIX_FEATURE_CONFIG` → `.helix/feature.yaml` → `config/feature/feature.yaml`. The SAME file is schema-compatible with `reporting.example.yaml` (§11.4.202) plus one addition, `research_doc_root` — it is passed straight through to `report_item.sh --config`.

Usage examples

Schedule a feature research effort (Type=Task, Status=Queued):

```
bash constitution/scripts/feature/schedule_feature_research.sh \  
  --description 'Add per-user configurable notification digest frequency (daily/weekly/off).'
```

Preview without writing anything (no delegate invocation, no queue append):

```
bash constitution/scripts/feature/schedule_feature_research.sh \  
  --description '...' --dry-run
```

Explicit title + machine-readable result:

```
bash constitution/scripts/feature/schedule_feature_research.sh \  
  --title 'Dark-mode theme toggle' \  
  --description 'Add a dark-mode theme toggle to the settings screen.' --json
```

Flag	Meaning
<code>--description</code> / <code>--description-file</code> / <code>stdin</code>	the feature description (in that precedence)
<code>--title</code>	optional; derived from the description's first sentence when omitted
<code>--severity</code> , <code>--reported-by</code>	optional; passed through to the delegate
<code>--config</code> , <code>--db</code>	optional overrides (else the search order above)
<code>--no-sync</code> , <code>--no-tracker</code>	skip step 2's sync / tracker-push (passed through to the delegate)
<code>--dry-run</code> , <code>--json</code>	preview / machine-readable result

Edge cases (all honest, none faked)

Situation	Behaviour
No config found	Exit 2 with the template path. The engine never guesses your DB path / research-doc root (§11.4.6).
The delegate (<code>re-report_item.sh</code>) is missing	Exit 3 — FEATURE never falls back to reimplementing item-creation itself.
Item creation fails	Exit 4 , propagated from the delegate; the FEATURE item was NOT created and NOT queued.
Sync command fails	Exit 5 , propagated from the delegate; the item IS created and IS queued (the DB is the SSoT) — re-run the sync.
Tracker credentials unset	Verdict <code>SKIP</code> , reason <code>credentials_absent: unset env: NAME ...</code> (via the delegate) — variable names only, never values (§11.4.10).
Tracker client not built	Verdict <code>SKIP</code> , reason <code>tracker_client_absent</code> (via the delegate).

A tracker `PASS` is recorded **only** after a real push command exits 0 — inherited from the delegate, never reimplemented here.

Internal behaviour

1. **Config** → this engine reads only its OWN key (`research_doc_root` , default `docs/research`) + `evidence_dir` ; every other key (`db` / `id_prefix` / `sync_command` / `trackers` / ...) is read independently by the DELEGATED `report_item.sh` from the SAME file.
2. **Title + destination slug** → derived from the description's first sentence (mirrors `report_item.sh`); the destination is `<research_doc_root>/<slug>_<UTC-timestamp>_<PID>/` — the trailing `$$` (this process's PID) is a uniquifier: two identical-title requests scheduled within the same second would otherwise

collide on the same (slug, seconds-resolution timestamp) key and record the SAME research-doc destination on two distinct items; the PID is already resolvable at zero cost and keeps the path human-readable.

3. **Work program** → the full §11.4.213 clause-(2) mandate ((a)–(k)) is rendered verbatim and embedded into the report text passed to the delegate, together with the destination path and the scheduling-not-synchronous execution model.
4. **Delegate** → `report_item.sh --kind task --title ... --report ... --scope ... --acceptance ... --config ... --json` — the SAME engine §11.4.202 uses, so the already-audited no-faked-push / decoupling / anti-bluff invariants are inherited, never duplicated.
5. **Queue** → append one row to `docs/requests/feature_queue.md` (created with a header + §11.4.44 revision line on first use). The title is the only user-influenced table cell; a literal `|` in it is escaped (`\|`) before the append so the row stays a well-formed 5-cell markdown row (an unescaped pipe would otherwise shift Destination / Scheduled / Status one column right) — the ITEM_ID / destination / timestamp / literal `Queued` cells are generated, never free-text, and need no escaping.
6. **Evidence** → `<evidence_dir>/<id>_<ts>/result.json` (this engine's own summary, embedding the delegate's own `report_item_result.json` , verbatim) (§11.4.5 / §11.4.69).

Decoupling (\$. . / \$. .)

The engine carries **zero project literals** and is inherited **by reference** (never copied). Every project-specific value is consumer-owned DATA. It also carries **zero duplicate item-creation / tracker-push logic** — that machinery lives exactly once, in `report_item.sh`. The `CM-FEATURE-DIRECTIVE` gate enforces both invariants (a hardcoded project literal, or a reimplemented tracker-push path, → gate FAILs).

Related

Path	Purpose
<code>constitution/actions/registry.yaml</code>	the <code>FEATURE</code> row + the shared §11.4.140 grammar
<code>constitution/scripts/action_prefix_lib.sh</code>	the §11.4.140 parser (all six forms)
<code>constitution/scripts/feature/feature.example.yaml</code>	the consumer config template
<code>constitution/scripts/gates/cm_feature_directive.sh</code>	the pre-build gate (+ its §1.1 mutation test)
<code>constitution/scripts/reporting/report_item.sh</code>	the DELEGATED §11.4.202 item-creation + full-sync engine
<code>constitution/scripts/workable-items/</code>	the DB SSoT binary (§11.4.93)

Last verified: 2026-07-16 — end-to-end against a real temp SQLite DB (via `sqlite3 <tmp>.db < schema.sql`): a dry-run preview, three real items created (`FTX-001` / `FTX-002` / `FTX-003`) with the correct `Type=Task` / `Status=Queued`, a configured tracker SKIPped honestly with `credentials_absent` , special characters (quotes / newlines / backslash) in the description round-tripped correctly through the DB + the JSON evidence, and `docs/requests/feature_queue.md` accumulated one row per scheduled item — all throwaway artefacts removed after the run.